

CSS

1. Introduction To CSS :

✓ Cascading Style sheet - 1997

✓ To describe the look and feel of the HTML Pages.

✓ Used with only HTML tags

✓ Supported by all web browsers.

```
<!DOCTYPE.html>
```

```
<html lang='en'>
```

```
<head>
```

```
<title> Nature beauty </title>
```

```
<link rel='stylesheet' type="text/css"
```

```
href="com.css">
```

```
</head>
```

```
<body>
```

```
<h1 align="center"> Nature </h1>
```

```
<div class='topbar'>
```

```
<ul>
```

```
<li> Home </li>
```

```
<li> Nature gallery </li>
```

```
<li> contact </li> </ul>
```

```
</div>
```

```
<div class="two"> <h3> Nature is beautiful  
gift given by God. It includes mountains,  
all living creatures such as birds, animals,
```

220
insects. If we live in good surroundings
our soul will be peaceful and happy.
</div>

Advantages :-

Flexibility to alter

Used to apply style to whole page
without changing the code.

Store in browser cache - used without
reloading.

Faster data downloads and decrease
the data transfer.

CSS Syntax :-

- Syntax consists of selector and declaration
part.
- Selector - p, div,
- Property - Color, height
- Value - green, 200px

Example

→ div { color : red ; height : 200px }

2. CSS Selectors :-

Basic Selectors

- Element Selector - HTML element
- Id Selector - #
- Class Selector - .

- universal selector - *

- Grouping selector - ,

< style

* {

font-family : "Courier New", Courier, monospace;

padding : 0;

}

body {

background-color : light blue;

width : 1000px;

padding-left : 10px;

padding-right : 10px;

margin : 0 auto;

}

h3, h2 {

color : blue;

font-size : 28px;

font-style : italic;

font-weight : bold;

}

h1 {

text-align : center;

color : navy;

p1 {

color : maroon;

font-style : italic;

font-family : Times to Roman

}

. center {

```
display : block;
margin-left : auto;
margin-right : auto;
}
```

```
h2, h3, h4 {
background-color : #green;
}
```

3. CSS Combinator Selectors :-

- Descendent Selectors - space
- Child Selectors >
- General Siblings Selectors ~
- Adjacent Siblings Selectors +

<style>

```
div p {
```

```
color : #green;
```

```
font-weight : bold;
```

```
}
```

```
ul > li {
```

```
display : inline;
```

```
}
```

} Descendent
Selectors

```
div > p {
```

```
color : #blue;
```

```
}
```

} Child selectors

```
#outer > .inner {
```

```
color : red;
```

```
} font-size : 20px;
```


h3 ~ p {

color : brown ;

font-size : 25px ;

} General Siblings
Selectors ~

h3 + ul {

color : gray ;

font-size : 16px ;

} Adjacent
Siblings
Selectors +

4. CSS Pseudo Selectors :-

- Pseudo class Selectors:-

Link , visited , active , hover , focus , first-child ,
last-child , nth-child , lang

- Pseudo Element Selectors :-

First-line , First-letter , before , after ,
Selection .

<style>

a: link {

color : yellow ;

}

a: visited {

color : #ff00ff ;

}

a: hover {

color : red ;

}

a: active {

color: # dodgerblue;

}

P. polar: hover {

background-color: # gold;

}

P. Polar: active {

background-color: # indianred;

}

emph: focus {

background-color: # coral;

}

li: first-child {

color: # white;

}

li: last-child {

color: # palevioletred;

}

li: nth-child(1) {

color: # royalblue;

}

P: first-child {

color: # yellowgreen;

}

P: last-child {

color: magenta;

}

text-indent : 5em;

font-weight : bold;

}

: lang (zuo) { quotes : '<' '>'; }

p :: first-letter {

font-family : Times to Roman;

font-size : 0.5em;

color : [] dark orange;

}

p-region :: first-line {

color : [] magenta;

font-variant : Small-Caps;

}

h3 :: before {

content : "Learn about";

}

^A
~~h3~~ : after {

content : "- climate";

}

h1 :: selection {

color : [] red;

}

text-indent

5. CSS Attribute Selectors []:-

- Attribute
- Attribute = Value
- Attribute ~ = Value
- Attribute | = Value
- Attribute 1 = Value

- attribute § = value
- Attribute * = value

```
Cclass {
```

```
color: #red;
```

```
}
```

```
B h3 Cclass, h4 Cclass {
```

```
color: #blue;
```

```
}
```

```
[title] {
```

```
color: #blueviolet;
```

```
font-weight: bold;
```

```
}
```

```
[title = 'animals'] {
```

```
background-color: #cornsilk;
```

```
}
```

```
[class = 'head']
```

```
{
```

```
background-color: #yellow;
```

```
}
```

```
input [type = 'text'] {
```

```
border: 3px solid: #blue;
```

```
}
```

```
[class ~ = 'region'] {
```

```
color: #chartreuse;
```

```
background-color: #darkblue;
```

```
}
```

(Copy)
(region)


```

[Class 1 = 'poc'] {
    font-weight : bold;
    font-style : italic;
    font-variant : small-caps;
    color : dark magenta;
}

```

(pot - not consider space)

```

[Class 1 = 'pol'] {
    font-family : cursive;
    font-size : 18px;
    background-color : darkseagreen;
}

```

(starting letter pol ->)

```

[Class $ = 'region'] {
    font-style : italic;
    font-size : 10px;
    background-color : pink;
}

```

ending word

```

[Class * = 'ola'] {
    color : deep pink;
    font-family : Times To Roman;
    font-size : 15px;
}

```

all font style

6. CSS Insertion Selector:-

4 ways to include style sheet in HTML document.

- Internal style sheet
- Inline style
- External style sheet

- Imported Style Sheet

Internal Style sheet:-

→ `<style> </style>`

→ Embedded within the head element of the HTML document.

Inline Style:-

→ Written in single line.

→ Placed inside the HTML element with style attribute.

`< style >`

`h3, h4 {`

`color: red;`

`padding: .5em;`

`}`

External Style sheet:-

→ Separate file of CSS with extension ".css".

→ Embedded within the head element of the HTML document

→ `<link>` tag is used to refer the external file.

→ Link attributes - href, type, rel, media.

<link type="text/css" href="include.css"
rel="stylesheet">

Import Style Sheet:-

- @import with URL is used to link the CSS style sheet - Inside <style> tag
- To import one style sheet into another CSS file..

Order of Application ::

1. Inline style
2. Internal style sheet
3. External style sheet
4. Browser default

@import "my design. css";

```
body {  
    background-color: rgb(0,255,255);  
    width: 200px;  
}
```

```
h1 {  
    color: green;  
    padding-top: 20px;  
}
```

```
#count {  
    font-size: 18px;  
    color: rgb(177,34,177);  
    font-weight: bold;  
}
```

Where do we use?

- Multiple pages - Common CSS ---> External style sheet

- One / two pages specific CSS ---> Internal style sheet

- Specific to element ---> Inline style

-> To include style sheet in other external CSS file ---> imported CSS.

7. CSS comments:-

- Single line comments /* */

- Multiline comments

/* */

..... */

Visual Studio Code - Ctrl + /
(Comment)

/* # Pad 1

color : maroon;

font-style : italic;

font-family : 'Times New Roman';

*/

/* Image aligned in centre */

1. Explanation of the code
2. For testing
3. Easy for editing
4. Browser didn't execute the part.

8. CSS colors :-

Basic ways of defining color in CSS

- Color keyword - red, yellow
- Hex color value = #ff0000, #ff00ff, #ffffff, #000000
- RGB color values - rgb(0, 0, 255)

h3, h2 {

color: #000080;

font-size: 20px;

}

h1 {

→ #ffffff

color: #ff0; ← short form

margin-left: auto;

}

h3, h4 {

color: rgb(0, 50, 225);

font-weight: bold;

}

9. CSS Background :-

CSS Background

To set the background of the element

- Background - color
- Background - image
- Background - repeat
- Background - attachment
- Background - position
- Background - size.

body {

width : 1000px;

margin : 0 auto;

background-color : # turquoise;

font-size : 20px;

line-height : 40px;

background-image : url('img/patterns.jpg');

}

CSS Background repeat

It specifies the repetition of the background image.

- Repeat - x
- Repeat - y
- No - Repeat

CSS Background Attachment

Background is fixed or scrolls with

The View Port

- Fixed

- Scroll

body {

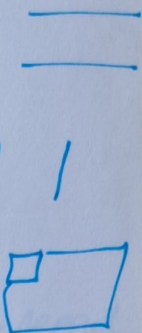
width: 1000px;

background-repeat: repeat-x;

background-repeat: repeat-y;

background-repeat: no-repeat;

}



body {

background-attachment: fixed;

background-attachment: scroll;

}

CSS Background Position:

It specifies the position of the background image.

- Default is top left

1. Horizontal

2. Vertical

default of second value is center

- Percentage

- Pixel-value

body {

width : 1000 px;

margin : 0 auto;

background - position : 50% 50% ;

background - position : top right ;

background - position : right ;

}

body {

background - size : 100 px 100 px ;

background - position : 68% 0% ;

}

h2, h3 {

background - image : url ("img / pattern2.jpg") ;

background - position : no-repeat ;

}

wild {

padding - left : 500 px ;

}

#

. cities {

background - image : url ("img / cities.jpg") ;

background - repeat : no-repeat ;

background - position : right ;

}

city {

width : 700 px ;

Background in short way :-

Background

1. Color
2. image
3. Repeat
4. Attachment
5. position

Background : red url("snow.jpg") no-repeat
Scroll top right.

10.

CSS Border :-

To set the border box of the element

- Border - style
- Border - color
- Border - width

CSS Border Sides :-

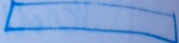
- Top
- Right
- Bottom
- Left
- Border - top - style
- Border - right - style
- Border - bottom - style
- Border - left - style

CSS Border Style:-

- none
- Solid
- Dotted
- Dashed
- Double
- ~~Groove~~ Groove
- Ridge
- Inset
- Outset
- Hidden

h4 # solid

{

border-style: solid; 

border-style: dashed; 

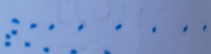
" " : Double; 

" " : groove; 

" " : Ridge; 

" " : Inset; 

" " : Outset; 

border-style: dotted; 

}

p. Style {

border-top-style: dotted;

border-right-style: double;

border-bottom-style: solid;

border-left-style: outset;

border-top-width: 5px;

border-right-width: 2px;

border-bottom-width : 5px;

border-left-width : 3px;

border-top-color : red;

border-right-color : green;

border-bottom-color : blue;

border-left-color : yellow;

border in short way :

1. width

2. style

3. color

Eg :

p {

border : 5px dashed yellow
}

h1 {

color : mediumvioletred;

text-align : center;

border : 5px dotted green; short form

}

p#p1 {

border-style : solid double;

border-width : 5px 3px 1px;

border-color : red indigo; red

border : 4px inset gray; red

}

P#P1 §

border-radius: 50px;

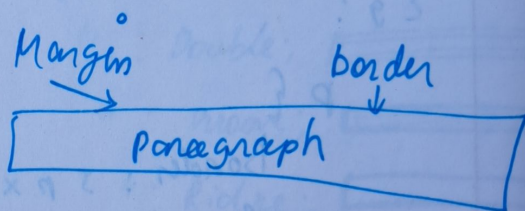
§

11.

CSS Margin :

To set the outer space of the element

- Margin-top
- Margin-right
- Margin-bottom
- Margin-left



h4 §

~~border~~

margin: 50px 20px 100px;

§

h4 §

margin-left: 20px;

margin-right: 20px;

margin-top: 10px;

margin-bottom: 5px;

§

Margin Values

- Length - px, cm, pt
- % - width of the element
- Auto - Automatically calculate by browser
- Inherit - margin value taken from its parent.

Negative margin:

- To overlap the element

h1 {

margin-bottom: -25px;

margin-top: 2%;

margin-right: 20px;

}

Margin Collapse:

- If the margin value of two consecutive element is different then large margin is taken into account.

→ If the margin value of two consecutive element is same then one margin value is taken into account.

Eg: p { margin-bottom: 30px; }

h4 { margin-top: 40px; } ⇒ 40px gap

between two elements.

b2, h3 {

margin-bottom: -20px;

}

#wild life {

margin-right: -50px;

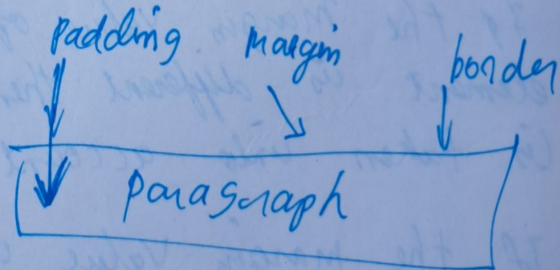
margin-left: 100px;

}

12. CSS padding:-

To set the space between the border and content of the element To set different values on each side of the element.

- padding-top
- padding-right
- padding-bottom
- padding-left



h4 {

padding: 30px;

padding: 25px 45px;

Padding Values:-

- length - px, cm, pt etc
- % - Width of the element

- inherit - padding values taken from its parent.

h3 {

padding-left: 4%;

padding-top: 1cm;

padding-bottom: 25pt;

padding: 50px;

line-height: 40px;

h4 {

padding-left: 200px;

padding: 25px 0px;

HEADING

13. CSS Width & Height :-

To set the width of the element

- width

To set the width of the element

- Height.

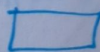
body {

width: 800px;

background-color: yellow;

margin: 0 auto;

width: 200px;



Container

Arrange

Object

width : auto ;

height : 40 px ;

color : red ;

background - color : cyan ;

width : 70 % ;

height : 30 px ;

padding - left : 15 px ;

padding - top : 10 px ;

⌋

⌈ place ⌈

height : 300 px ;

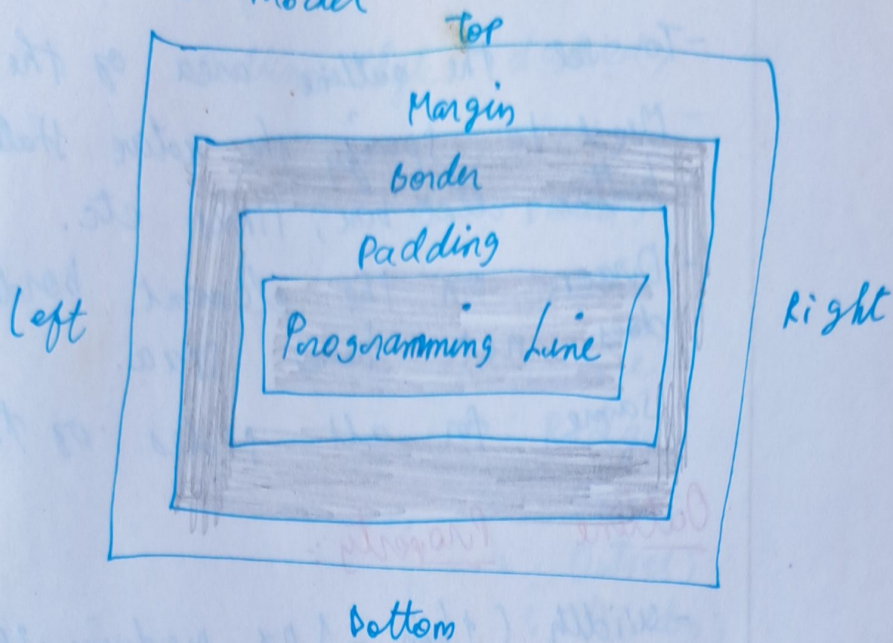
width : 300 px ;

⌋

14. CSS Box Model :-

- > Content area surrounded by one or more rectangular boxes
- > Used to design the element layout of the HTML document
- > Element are treated as rectangular box.
- > Container comprises of margin, border, padding and content.

CSS box Model



box 1 {

background-color : # crimson ;
border : 25 px solid # yellow ;
padding : 25 px ;
margin : 25 px ;
width : 500 px ;
height : 400 px ;

CSS Box Size:-

Box width = content width + padding-left - padding-right + margin-left + margin-right + border-left + border-right.

Box height = content height + padding-top + padding-bottom + margin-top + margin-bottom + border-top + border-bottom.

15.

CSS Outline :-

- To set the outline area of the element
- Used to specify the active states of the button, text box, links etc.
- Drawn on the element border so it does not take space.
- Same for all sides of the elements.

Outline Property :-

- width (thin - 1px, medium - 3px, thick - 5px, values)
- style
- color
- offset

Outline Over border :-

- Takes no space to draw outline on the element.
- Same for all sides - not different values for different sides.
- do not have to be in rectangular

Outline Style :-

- | | |
|----------|----------|
| - none | - Groove |
| - Solid | - Ridge |
| - Dotted | - Inset |
| - Dashed | - Outset |
| - Double | - Hidden |

→ any name
h4 # dash {

Outline Style : dashed;

" " : solid;

" " : dotted;

" " : Double;

" " : Groove;

" " : Ridge;

" " : Inset;

" " : Outset;

" " : Hidden;

}

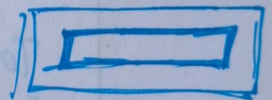
P. Pstyle {

→ thin, thick, medium

Outline : 5px solid green;

border : 4px ~~solid~~ dashed red;

Outline-offset : 10px



16. CSS Text :-

- | | |
|---------------------|--------------------|
| - color | - Text - Shadow |
| - Text - Indent | - Letter - Spacing |
| - Text - align | - Word - Spacing |
| - Text - decoration | - Line - height |
| - Text - transform | - White - Space. |

CSS Color

It specifies the color of the text

- Color name (red, blue, ...)
- RGB Value (rgb(255, 34, 44))

- Hex value (#ff00ff)

CSS text-indent

To set the indentation of the first line of the text

- length values - px, pt, cm etc
- Percentage (%)

```
h1 {
```

```
  width: 100%;
```

```
  color: #ff00ff;
```

```
  color: yellow;
```

```
  text-indent: 40px;
```

```
}
```

CSS text-align

To set the horizontal alignment of the text

- Left, right, center, justify

CSS line-height

To set the space between the lines

- length values
- Percentages

```
p {
```

```
  line-height: 100%;
```

```
}
```


CSS text-decoration

It specifies the decoration of the text
- underline, overline, line-through, none.

CSS text-transform

To set the cases for the text
- uppercase, lowercase, capitalize

h4 {

text-decoration: underline; link

text-decoration: overline; text

text-decoration: none;

text-decoration: line-through; ~~text~~

}

h4 {

text-transform: uppercase;

text-transform: lowercase;

text-transform: none;

}

CSS text-shadow:

It adds the shadow effect to the text
- Horizontal, vertical, blur with color.

CSS letter - spacing :-

To set the space between each character of the text - Normal, length values, Negative values.

h3 {

text-shadow: 2px 2px hotpink;

text-shadow: 2px 2px 10px hotpink;

↳ blur effect

text-shadow: 2px 2px 10px yellow,

2px 10px 10px pink;

h3 {

Letter-spacing: 10pt;

Letter-spacing: 2px;

Letter-spacing: 10%;

}

CSS WORD - Spacing

To set the space between each word of the text.

- Normal, length values, negative values

CSS White - Space

To set the white space of the text

- Normal, Pre, nowrap

h4 {

Word - Spacing : 20px;

Word - Spacing : 10pt;

Word - Spacing : -10px;

Word - Spacing : normal;

}

h4 {

white - space : pre;

white - space : Normal;

white - space : nowrap;

}

17.

CSS Font :

To set the styling font of the text

- Font - Family

- Font - style

- Font - weight
- Font - size
- Font - Variant

Font - family :-

- Serif
- Sans - serif
- Mono Space
- Cursive
- fantasy

h4 {

font - family : Sans - serif;

" " : serif;

" " : Mono Space;

" " : Cursive;

" " : fantasy;

}

call off
Verdana, Geneva, Tahoma,
Sans - serif;

Font - style :-

It specifies the font style of the text of the element

- normal, italic, Oblique

Font - variant :-

To display the text in small capital letters

→ Small - caps

h4 {

font - style : italic;

" " : oblique;

" " : ~~normal~~ normal;

}

h4 {

font - variant : small-caps;

}

font - weight :-

To set the boldness of the text

- Values (100 --- 900)

- Normal - 400

- Bold - 700

- Bolder

- lighter

h4 {

font - weight : 700;

font - weight : normal;

font - weight : bolder;

}

18. CSS Font - Size :-

To set the size of the font for the text content of an element

- Pixel values
- Keywords -- large (18), small (13), medium (16)
x-large (24), xx-large (32), x-small (10)
xx-small (9), larger, smaller
- Percentage
- Em - 1em = font size of the parent element (16 px)
- Vw or vh --- Viewport width or viewport height --- 2vh = 2% of viewport height

```
h4 { font-size: 40%; → 40px;  
      font-size: small;  
      " " : large;  
      " " : medium;  
      " " : x-small;  
      " " : xx-small;  
      " " : xx-large;  
      " " : x-large;  
      " " : 3em;  
}
```

Font shorthand :-

To set the font properly in one line

Font: style, font-variant, font-weight,
font-size, font-family

Eg:

h1 { font: italic small-caps bold 32px orial; }

p { font: italic 18px courier; }

h4 {

font: italic bold 30px Verdana;

19.

CSS Links:

Links - Connection from one page to other web pages in the same tab or in new tab

States of link

- link - unvisited
- Visited -- Visited by user
- hover -- When the mouse over the link.
- active -- Currently clicked by user.

CSS default link styles

- Links are underlined
- Hover - hand icon
- Unvisited link - blue color
- Visited link - purple color
- Active link - red color

CSS link Properties

- Color
- Background color
- Text-decoration
- Font-family

- Border

a: link {

color: # hotpink;

}

a: visited {

color: # green;

}

a: hover {

color: # indianred;

}

a: active {

color: # cyan;

}

a: link {

text-decoration: none;

"

": underlined;

"

": overline;

padding: 10px; 5px

border: 2px solid;

border-radius: 10px; 20px

}

Qo. CSS List:-

Lists - Ordered list, Unordered list

Lists style : border width - color

list - style - type $\Rightarrow$ To display different shapes and appearances of marker

list - style - image $\Rightarrow$ Images can be used instead of marker

list - style - position $\Rightarrow$ To control the position of the marker.

Lists - Marker Type:-

List - style - type $\Rightarrow$ Used to change list's marker type unordered lists

- disc, circle, square, none

Ordered lists

- upper - roman, lower - alpha, lower - roman, decimal, decimal - leading - zero, none.

Lists - Position :-

Lists - style - position $\Rightarrow$ Used to control the position of the marker.

- Inside - marker will be displayed within the list item
- Outside - marker will be displayed outside the list item (default)

Lists - image :-

List - style - image: url ('img/11con.png');

Lists - short hand:

This property is used to set all Properties in one line.

- type
- position
- image

ul. ch {

list-style-type: square;

list-style-position: inside;

}

ol. phy {

list-style-type: upper-roman

list-style-type: decimal-leading

-Zero;

list-style-position: inside;

"

"

Outside;

}

ol. che {

list-style-type: none;

list-style-image: url('icon!.png');

}

ul. recall {

list-style-type: none;
{

ul. recall li::before {

content: ' ';
display: inline-block;
height: 35px;
width: 35px;
background-size: 35px
background-image: url('flower.png');
margin-right: 10px;
padding-bottom: 10px;
vertical-align: middle;
background-repeat: no-repeat;

ul. ch {

list-style: circle outside;
{

91. CSS Lists - Menu :-

- Navigation Menu
- Drop down Menu

<style>

body {

background-color: # teal;

font-family: cursive;

margin: 0;

h1 {

color : # yellow;

text-align : center;

padding : 10 px;

ul {

background-color : # skyblue;

list-style : none;

}

ul li {

background-color : # skyblue;

display : inline-block;

width : 150 px;

padding : 10 px;

}

ul li a {

text-decoration : none;

color : # brown;

font-size : 20px;

font-weight : 700;

}

ul li a :hover {

color : # white;

background-color : # blue;

padding : 10px;

}

<body>

<h1> Programming Line </h1>

<div class = "nav">

 MENU

 MENU

 MENU

~~ MENU ~~

</div>

</body>

</html>

↑ Even

ul {

list-style : none;

}

ul li {

width : 150px;

text-align : center;

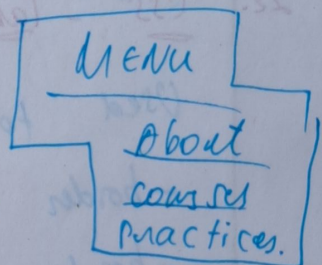
}

ul li a {

text-decoration : none;

display : block;

background-color : lightpink;



```
padding: 10px;
```

```
}
```

```
ul li ul {
```

```
display: none;
```

```
}
```

```
ul li: hover ul {
```

```
display: block;
```

```
}
```

```
- nav
```

```
{
```

```
padding: 500px;
```

```
}
```

22. CSS - Tables :-

Used to style the HTML tables

- border : width, color

- border - collapse; collapse, separate

- border - spacing : values

```
<body>
```

```
<h1> Employee's Information Table </h1>
```

```
<table class = "emp" style = border = "1" >
```

```
<tr> <th> Name </th> <th> Designation </th>  
      <th> Exp. Yrs </th> </tr>
```

```
<tr> <td> Rakesh </td> <td> Manager </td> <td> 4 </td>  
      </tr>
```

```
<tr> <td> Meera </td> <td> Employee </td> <td> 1  
      </td> </tr>
```


</table>

body {

background-color: antiquewhite;

color: blue;

font-size: 20px;

font-family: "sans serif";

width: 800px;

margin: 0 auto;

}

table, th, td {

border: 2px solid red;

}

h1 {

color: sienna;

font-align: center;

border-collapse: collapse; → separate;

table {

border-spacing: 10px;

CSS - Tables :-

- Caption - side : top, bottom
- empty - cells : show, hide
- table layout : auto, fixed

<body>

<caption> Table 1.1 Employee's Information </caption>
<table>

<tr> <td> Rakesh </td> <td> Employee </td> </tr>
</tr>

</table>

table, th, td {

caption - side : bottom;

empty cells : hide;

" " : show;

text-align : right;

width : 100px;

table-layout : fixed;

" " : auto;

}

CSS - Table Text alignment :-

- Table size : width, height

- Horizontal alignment : text-align

- Vertical alignment : vertical-align

- Text space : padding

table {

width : 80%;

height : 25px;

padding : 5px;

vertical-align : middle;

" " : top;

" : bottom ;
back ground - color : lightgreen ;
}

23. CSS Display :-

Used to change the default display of HTML element on the web page.

- block
- inline
- inline - block
- none

h4 {

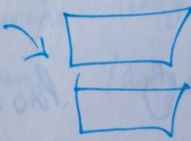
margin - top : 20px ;

Display : inline ;

" : inline - block ;

" : block ;

" : none ;



}

display : visibility ;

a : Visited {

color : lightgreen ;

display : block ;

}

display : none ;

hide

a : : hover {

display : block ;

}

a : : li {

display : inline ;

height : 50px ;

width : 50px ;

}

| |
|---------|
| home |
| Product |
| |
| |

background: attachment: fixed



Home



Product



Service



about as.



contact as